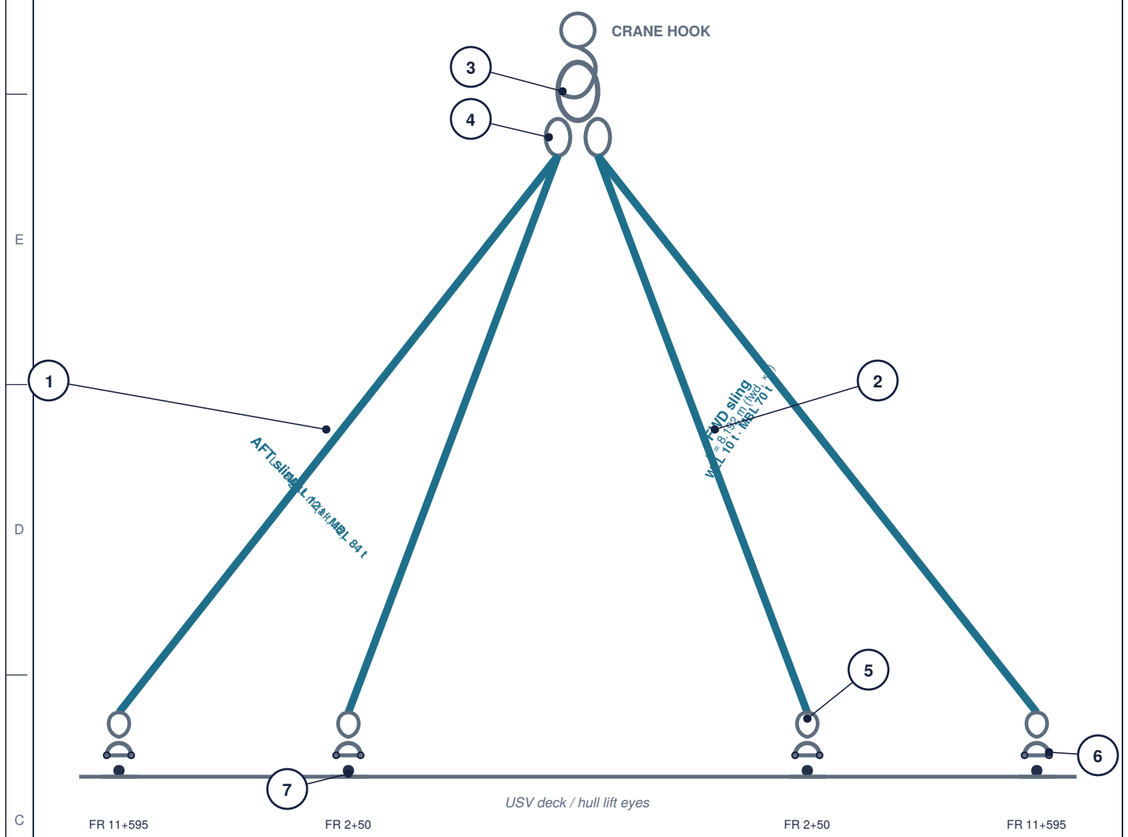
# LOAD CASE — 22 TON · PORT RECOVERY (0 wave → deck), LCG = 5.107 [m]

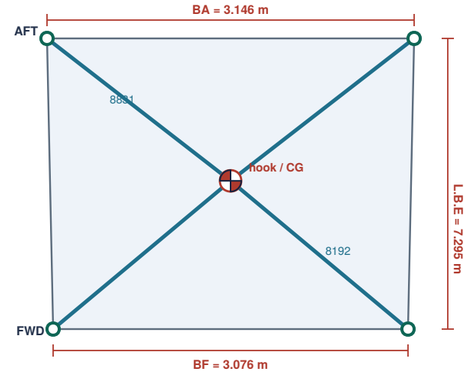
LIFTING SET FOR 22 t · DAF 1.15 (sheltered / inshore recovery) · parameters & constraints per Ginton dwg 287-235

## VIEW A — FRONT ELEVATION (bow-on) · 4-leg bridle to single hook

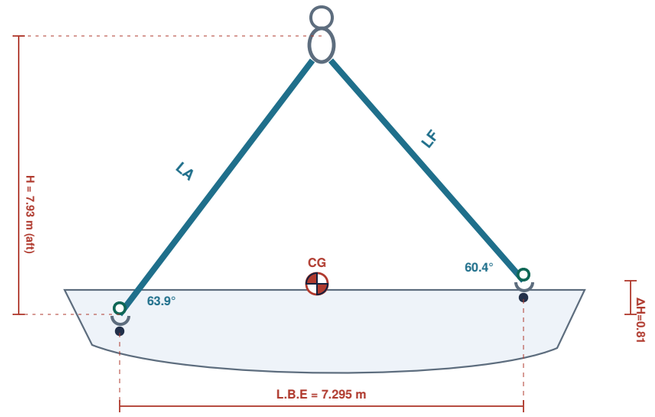



*Soft slings choked/eyed to Crosby end fittings · matched pairs · level lift ≤ 2°*

## VIEW B — PLAN (lift-point pattern)




## VIEW C — SIDE ELEVATION (profile)




*AFT ← → FWD · aft legs longer to hold level lift under CoG offset & ΔH*

## DETAIL A — CROSBY END-FITTING STACK + LOADS

| | | |
|---|---|---|
| 3 | **Master link assembly** — Crosby A-345W 1.50 / 1014807 | WLL 27.7 t / MBL 138.5 t |
| 4 | **Master link — sub/child** — Crosby A-345 child | WLL 27.7 t |
| 5 | **HP connector S-237 Fr.25** — Crosby 1020722 · "Hammerlok" | WLL 11.34 t / MBL 56.7 t |
| 6 | **Anchor shackle G-209A 1"** — Crosby 1017582 · bolt type | WLL 12.5 t / MBL 62.5 t |

*Stack indicative · all Crosby alloy, certified · DF: shackle & S-237 = 5:1*

## DETAIL B — HULL-EYE / SLING BEARING (§8.2)

| | |
|---|---|
| w ≥ 1.25·d + 25 mm | [§8.2.1] |
| D/d ≥ 4 → eff. ≥ 0.75 | [§5.12] |
| pin-hole Ø = pin +2 mm | [§8.4.1] |
| ovalises → radiused seat | [§8.2.2] |
| Secutex SF-2 at bearing | — |
| 100% visual+MPI welds | [§8.6.1] |

| ITEM | PART NUMBER | DESCRIPTION | QTY | WLL / SWL | MBL |
|---|---|---|---|---|---|
| 1 | RS-EL-SEAGULL_LIFT-10001-0 | 8.831 m · 12 t roundsling, AFT (SpanSet Magnum) | 2 | 12 t | 84 t (7:1) |
| 2 | RS-EL-SEAGULL_LIFT-10002-0 | 8.192 m · 10 t roundsling, FWD (SpanSet Magnum) | 2 | 10 t | 70 t (7:1) |
| 3 | CROSBY A-345W 1.50 / 1014807-MASTER | 27.7 t master link | 1 | 27.7 t | 138.5 t (5:1)* |
| 4 | CROSBY A-345W 1.50 / 1014807-CHILD | 27.7 t master link — sub/child | 2 | 27.7 t | 138.5 t (5:1)* |
| 5 | CROSBY 237_25 / 1020722 | Hammerlok sling-saver HP connector S-237 Fr.25 | 4 | 11.34 t | 56.7 t (5:1) |
| 6 | CROSBY G-209A_1.00 / 1017582 | Bolt-type alloy anchor shackle, 1" | 4 | 12.5 t | 62.5 t (5:1) |
| 7 | SPANSET SECUTEX SF-2 | Protective sleeve at hull-eye bearing | 8 | — | — |

*WLL = Working Load Limit/SWL · MBL = Minimum Breaking Load · *master-link MBL per Crosby cert. (assumed 5:1). Lengths in mm; nominal (actual).*

## DESIGN LOADS — GL Noble Denton 0027/ND Rev 10

| | |
|---|---|
| Design weight (Ginton 602M) | 22.0 t (21.76 t) |
| Dynamic hook (×DAF 1.15) | 25.0 t ≤ master link 27.7 t (90% · SF 5.5:1) |
| AFT sling — static / design (×DAF×SKL) | 6.21 t / 8.93 t ≤ WLL 12 t (74%) |
| FWD sling — static / design | 6.10 t / 8.77 t ≤ WLL 10 t (88%) |
| Shackle G-209A | MBL/3 = 20.8 t ≥ 8.93 t · WLL 12.5 ≥ 6.21 |
| Factors | DAF 1.15 (sheltered/inshore) · SKL 1.25 · soft-sling SF 7:1 |

## NOTES

SCOPE: recovery of floating USV from sea AT PORT, 0 wave height,
lifted to deck (sheltered) → DAF 1.15, NOT 1.30 (open-sea).
Hook height (computed, Ginton): 7.93 m above aft eyes / 7.12 m fwd.
Open-sea lift would need DAF 1.30 + uprated master link (≈ 31 t).
No personnel aboard; equal fuel fill if <100%; matched pairs.

**אורכי רצועות □ חוזק רצועות □□□□□□□□□**
**שאקל □□□□-□ ועומס □ טבעת מסטר □□□-□ ועומס**

## RIGGING DESIGN — AI-ASSISTED (A. SAPIR)

| | | | |
|---|---|---|---|
| TITLE: LIFTING SET 22 t · PORT | | BASIS: Ginton 287-235 · GL 0027/ND | |
| DRG No.: STRAP-ASSY-22T-P | REV: 4 | SIZE: A3 | SHEET: 1 / 1 |
| DRAWN: A. SAPIR | DATE: 28-05-26 | SCALE: NTS | UNITS: mm / t |
| PROJECTION: 1st angle | | SYMBOL | |